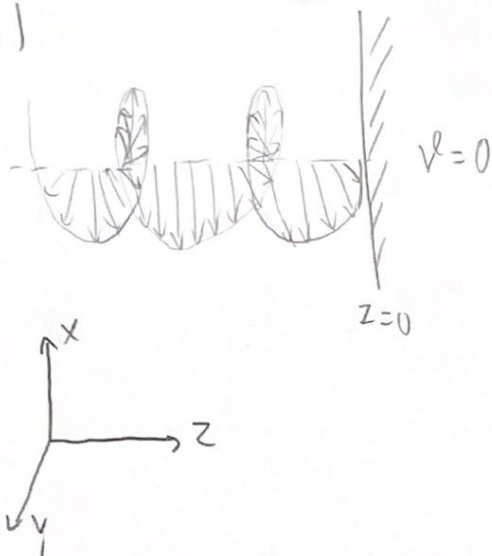


4)



In the last homework we have found out that right circularly polarized waves were of the form:

$$\vec{f}_R = f_x^0 \sin(kz - \omega t + \phi_x) \hat{i} + f_y^0 \cos(kz - \omega t + \phi_x) \hat{j}$$

and left circularly polarized waves were of the form:

$$\vec{f}_L = f_x^0 \sin(kz - \omega t + \phi_x) \hat{i} - f_y^0 \cos(kz - \omega t + \phi_x) \hat{j}$$

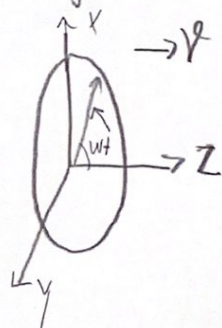
★ So the main idea is we can think of the circularly polarized waves as a combination of sinusoidal waves oscillating in perpendicular planes and they are out of phase by $\pi/2$.

We first send a right circularly polarized wave moving to the right. That is:

$$\vec{f}_i = f_x^0 \sin(kz - \omega t + \phi_x) \hat{i} + f_y^0 \cos(kz - \omega t + \phi_x) \hat{j}$$

$$\text{we have } (\hat{i}) \times (\hat{j}) = (\hat{k})$$

↳ cross product is right handed



We can think of the wall as a medium with infinite mass density or $v_{\text{wall}} = 0$

Utilizing the result we have found in questions 2 & 3:

$$f_R = f_i \left(\frac{v_R - v_L}{v_R + v_L} \right) \quad f_T = f_i \frac{2v_R}{v_R + v_L}$$

$$\lim_{v_R \rightarrow 0} f_R = -f_i, \quad \lim_{v_R \rightarrow 0} f_T = 0$$

Then sinusoidal wave forms will change direction and their amplitudes will be reversed:

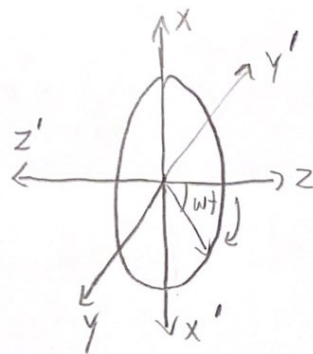
$$\vec{f}_R = -f_x^0 \sin(-kz - \omega t + \phi_x) \hat{i} - f_y^0 \cos(-kz - \omega t + \phi_x) \hat{j}$$

$$\text{OR } \vec{f}_R = f_x^0 \sin(kz' - \omega t + \phi_x) \hat{i}' - f_y^0 \cos(kz' - \omega t + \phi_x) \hat{j}'$$

$$\text{where } \hat{i}' = -\hat{i}, \hat{j}' = -\hat{j}, \hat{k}' = -\hat{k}$$

$$\text{Now we have } \hat{i}' \times \hat{j}' = -\hat{k}'$$

which shows that the reflected wave is left circularly polarized.



It can also be realized with the following perspective:

$$\vec{f}_R = -f_x^0 [-\sin(kz + \omega t - \phi_x) \hat{i}] - f_y^0 [\cos(kz + \omega t - \phi_x) \hat{j}]$$

$$= f_x^0 \sin(kz + \omega t - \phi_x) \hat{i} - f_y^0 \cos(kz + \omega t - \phi_x) \hat{j}$$

which is actually the form of the left circularly polarized wave we're familiar with

5) Consider the Lagrangian $L = \frac{1}{2} m \dot{q}^2 - \frac{1}{2} m \omega^2 q^2$

Define $p \equiv \frac{\partial L}{\partial \dot{q}}$, then $p = m \dot{q}$

$H \equiv p \dot{q} - L(q, \dot{q}) \rightarrow$ Although it looks like a function of $\dot{q}$, we can replace $\dot{q}$ with $\dot{q} = \frac{p}{m}$

$$H \equiv p \frac{p}{m} - \frac{1}{2} m \dot{q}^2 + \frac{1}{2} m \omega^2 q^2 = \frac{p^2}{m} - \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2 = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2,$$

One can show ^{that} $H \equiv p \dot{q} - L$ doesn't explicitly depend on $\dot{q}$ if

$$\frac{\partial H}{\partial \dot{q}} = 0 \Rightarrow \frac{\partial H}{\partial q} = \frac{\partial}{\partial \dot{q}} (p \dot{q} - L(q, \dot{q})) = p - \underbrace{\frac{\partial L}{\partial \dot{q}}}_{p \text{ by definition}} = 0,$$

Now if we know that Euler-Lagrange equations hold:

$$\frac{d}{dt} \left(\frac{\partial L(q, \dot{q})}{\partial \dot{q}} \right) = \frac{\partial L(q, \dot{q})}{\partial q} \quad \text{and} \quad L(q, \dot{q}) = p \dot{q} - H(q, p)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = \frac{d}{dt} \left[p - \underbrace{\frac{\partial H(q, p)}{\partial \dot{q}}}_{0 \text{ since independent of } \dot{q}} \right] = \frac{dp}{dt} = \frac{\partial(p \dot{q})}{\partial q} - \frac{\partial H}{\partial q}$$

Then we have $\dot{p} = - \frac{\partial H(q, p)}{\partial q}$

$$\text{Also } \frac{\partial}{\partial p} \left(\underbrace{L(q, \dot{q})}_{0 \text{ since } L} \right) = \frac{\partial}{\partial p} (p \dot{q} - H(q, p)) \Rightarrow 0 = \dot{q} - \frac{\partial H}{\partial p}$$

is a function of $q, \dot{q}$ only. It doesn't explicitly depend on p

$$\Downarrow \\ \dot{q} = \frac{\partial H(q, p)}{\partial p}$$

* We can define the state of a system by knowing q, p instead of $q, \dot{q}$. Then one can define the 2 vector $\begin{pmatrix} q \\ p \end{pmatrix}$

$$\left. \begin{aligned} \frac{dq}{dt} &= 0 + \frac{\partial H}{\partial p} \\ \frac{dp}{dt} &= -\frac{\partial H}{\partial q} + 0 \end{aligned} \right\} \frac{d}{dt} \begin{pmatrix} q \\ p \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial q} \\ \frac{\partial}{\partial p} \end{pmatrix} H$$

partial derivative operator acting on H

$$\left. \begin{aligned} \frac{dH}{dt} &= \frac{\partial H}{\partial q} \frac{dq}{dt} + \frac{\partial H}{\partial p} \frac{dp}{dt} \\ &= \dot{q} \frac{\partial H}{\partial q} + \dot{p} \frac{\partial H}{\partial p} \end{aligned} \right\} \begin{aligned} &\text{since } H = H(p, q) \\ &\text{If } H \text{ was } H(p, q, t): \\ &\frac{dH}{dt} = \frac{\partial H}{\partial q} \frac{dq}{dt} + \frac{\partial H}{\partial p} \frac{dp}{dt} + \frac{\partial H}{\partial t}, \end{aligned}$$

* If the Lagrangian describes a closed system (i.e. not interacting with anything outside), it can not depend explicitly on time. That is

$$\frac{\partial L}{\partial t} = 0 \Rightarrow -\frac{dL}{dt} = \frac{\partial L}{\partial q} \dot{q} + \frac{\partial L}{\partial \dot{q}} \ddot{q} =$$

Assuming Euler-Lagrange equations hold:

$$\frac{dL}{dt} = \dot{q} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) + \frac{\partial L}{\partial \dot{q}} \ddot{q} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \dot{q} \right)$$

$$0 = \frac{d}{dt} \left(\underbrace{\frac{\partial L}{\partial \dot{q}} \dot{q}}_p - L \right) = \frac{d}{dt} (p\dot{q} - L)$$

But the term in parenthesis is just the Hamiltonian!
Then we can conclude that H is constant in time. Therefore we have

$$\frac{dH}{dt} = \dot{q} \frac{\partial H}{\partial q} + \dot{p} \frac{\partial H}{\partial p} = \frac{\partial H}{\partial p} \frac{\partial H}{\partial q} - \frac{\partial H}{\partial q} \frac{\partial H}{\partial p} = 0 \quad //$$

By recalling $\dot{q} = \frac{\partial H}{\partial p}$, $\dot{p} = -\frac{\partial H}{\partial q}$

Now define $A(p, q, t)$ & $B(p, q, t)$

A Poisson Bracket is defined as: $\{A, B\}_{PB} \equiv \frac{\partial A}{\partial q} \frac{\partial B}{\partial p} - \frac{\partial A}{\partial p} \frac{\partial B}{\partial q}$

$$\begin{aligned} \frac{dA}{dt} &= \frac{\partial A}{\partial p} \frac{dp}{dt} + \frac{\partial A}{\partial q} \frac{dq}{dt} + \frac{\partial A}{\partial t} \\ &= \frac{\partial A}{\partial q} \frac{\partial H}{\partial p} - \frac{\partial A}{\partial p} \frac{\partial H}{\partial q} + \frac{\partial A}{\partial t} = \{A, H\} + \frac{\partial A}{\partial t} // \end{aligned}$$

$$\{B, A\} = \frac{\partial B}{\partial q} \frac{\partial A}{\partial p} - \frac{\partial B}{\partial p} \frac{\partial A}{\partial q} = - \left[\frac{\partial A}{\partial q} \frac{\partial B}{\partial p} - \frac{\partial A}{\partial p} \frac{\partial B}{\partial q} \right] = - \{A, B\} //$$

$$\begin{aligned} \{A, \{B, C\}\} &= \frac{\partial A}{\partial q} \frac{\partial}{\partial p} \left(\frac{\partial B}{\partial q} \frac{\partial C}{\partial p} - \frac{\partial B}{\partial p} \frac{\partial C}{\partial q} \right) - \frac{\partial A}{\partial p} \frac{\partial}{\partial q} \left(\frac{\partial B}{\partial q} \frac{\partial C}{\partial p} - \frac{\partial B}{\partial p} \frac{\partial C}{\partial q} \right) \\ &= \frac{\partial A}{\partial q} \left(\frac{\partial^2 B}{\partial p \partial q} \frac{\partial C}{\partial p} + \frac{\partial B}{\partial q} \frac{\partial^2 C}{\partial p^2} - \frac{\partial^2 B}{\partial p^2} \frac{\partial C}{\partial q} - \frac{\partial B}{\partial p} \frac{\partial^2 C}{\partial p \partial q} \right) \\ &\quad - \frac{\partial A}{\partial p} \left(\frac{\partial^2 B}{\partial q^2} \frac{\partial C}{\partial p} + \frac{\partial B}{\partial q} \frac{\partial^2 C}{\partial p \partial q} - \frac{\partial^2 B}{\partial p \partial q} \frac{\partial C}{\partial q} - \frac{\partial B}{\partial p} \frac{\partial^2 C}{\partial q^2} \right) \\ &= A_q B_{pq} C_p + A_q B_q C_{pp} - A_q B_{pp} C_q - A_q B_p C_{pq} - A_p B_{qq} C_p - A_p B_q C_{pq} + A_p B_{pq} C_q \\ &\quad + A_p B_p C_{qq} \end{aligned}$$

Similarly:

$$\begin{aligned} \{C, \{A, B\}\} &= C_q A_{pq} B_p + C_q A_q B_{pp} - C_q A_{pp} B_q - C_q A_p B_{pq} - C_p A_{qq} B_p - C_p A_q B_{pq} + C_p A_{pq} B_q \\ &\quad + C_p A_p B_{qq} \\ \{B, \{C, A\}\} &= B_q C_{pq} A_p + B_q C_q A_{pp} - B_q C_{pp} A_q - B_q C_p A_{pq} - B_p C_{qq} A_p - B_p C_q A_{pq} + B_p C_{pq} A_q + B_p C_p A_{qq} \end{aligned}$$

If we add them up all the terms will cancel:

$$\{A, \{B, C\}\} + \{C, \{A, B\}\} + \{B, \{C, A\}\} = 0$$

This property is called Jacobi identity and is valid for other objects having circular symmetry such as the cross product.

Another way of realizing this property is letting $A=B$

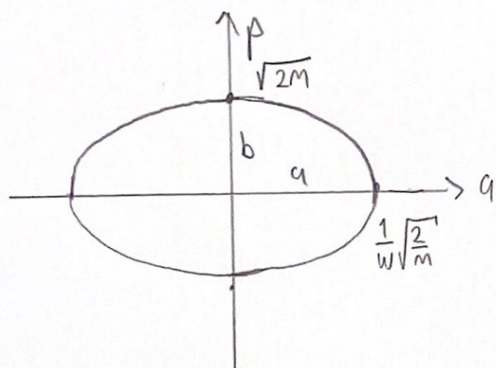
Since $\{A, B\} = \{A, A\} = 0$, we will have.

$$\{A, \{A, C\}\} + \{A, \{C, A\}\} = \{A, \{A, C\}\} - \{A, \{A, C\}\} = 0$$

Since setting $B=C$ or $A=C$ would give the same result, one can conclude that, it should generally be equal to 0.

Now, we've found out that H must be constant in time. Then:

$H = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2 : \text{const}$ is represented by an ellipse on the (p, q) plane



minor axis: $b = \sqrt{2m}$

major axis: $a = \frac{1}{w} \sqrt{\frac{2}{m}}$

6)

We're given the equation of motion of a system: $\ddot{q} + q - q^3 = 0$

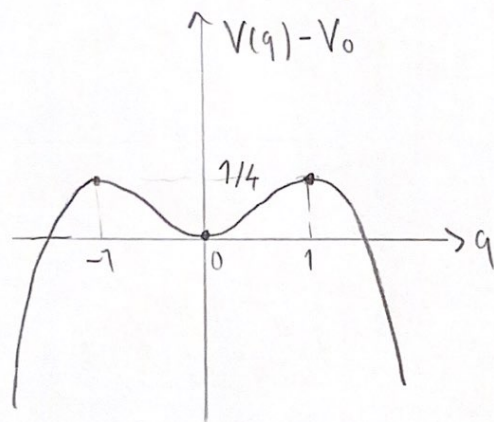
Presume it comes from a Lagrangian as such:

$$L = \frac{\dot{q}^2}{2} - V(q)$$

If Euler-Lagrange equations hold we will have:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = \frac{\partial L}{\partial q} \Rightarrow \ddot{q} = - \frac{\partial V}{\partial q}$$

$$\text{Then } \frac{\partial V}{\partial q} = q - q^3 \Rightarrow V(q) = \frac{q^2}{2} - \frac{q^4}{4} + V_0$$



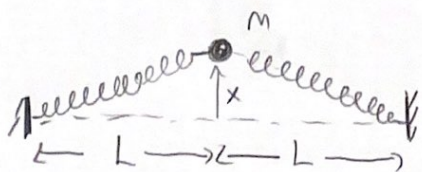
→ This potential is symmetric around $q=0$

→ It has a stable point at $q=0$

→ It has 2 unstable points at:

$$q = +1, q = -1$$

→ This potential is realizable with a system where an object of mass m is connected with 2 springs of equal length from either side:



Assume unstretched length of the springs ≈ 0 . Then when the ball is displaced by a small amount x in the vertical direction, springs are stretched by:

$$\Delta x = \sqrt{L^2 + x^2} = L \left(1 + \frac{x^2}{L^2} \right)^{1/2}$$

If x is small we can make the approximation $(1+\alpha)^n \approx 1+n\alpha$

$$\Delta x \approx L \left(1 + \frac{1}{2} \frac{x^2}{L^2} \right)$$

Then potential energy stored in the springs due to their elastic structure is basically:

$$V(x) = 2 \frac{1}{2} k (\Delta x)^2 = k L^2 \left(1 + \frac{1}{2} \frac{x^2}{L^2} \right)^2 = k L^2 \left(1 + \frac{x^2}{L^2} + \frac{x^4}{4L^4} \right) \\ = k \left(L^2 + x^2 + \frac{x^4}{L^2} \right)$$

→ This yields a "quartic potential" just as in our Lagrangian.

* This type of quartic potential is also commonly encountered in "double well problem" in quantum mechanics.

→ Propose a solution $q(t) = \pm \tanh\left(\frac{t}{\sqrt{2}}\right)$

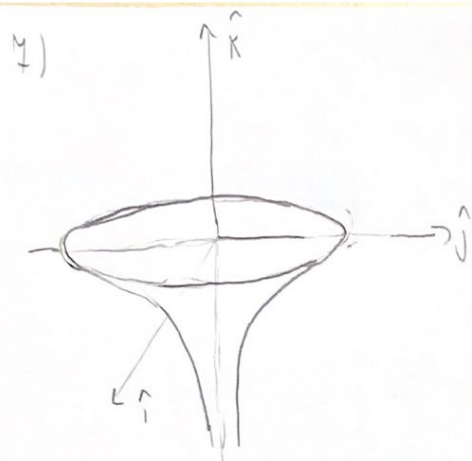
$$\dot{q}(t) = \pm \operatorname{sech}^2\left(\frac{t}{\sqrt{2}}\right) \cdot \frac{1}{\sqrt{2}} \quad , \quad \ddot{q}(t) = \mp 2 \tanh\left(\frac{t}{\sqrt{2}}\right) \operatorname{sech}^2\left(\frac{t}{\sqrt{2}}\right) \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \\ = \mp \tanh\left(\frac{t}{\sqrt{2}}\right) \operatorname{sech}^2\left(\frac{t}{\sqrt{2}}\right)$$

$$\mp \tanh\left(\frac{t}{\sqrt{2}}\right) \operatorname{sech}^2\left(\frac{t}{\sqrt{2}}\right) \pm \tanh\left(\frac{t}{\sqrt{2}}\right) \mp \tanh^3\left(\frac{t}{\sqrt{2}}\right)$$

$$= \tanh\left(\frac{t}{\sqrt{2}}\right) \left[\mp \operatorname{sech}^2\left(\frac{t}{\sqrt{2}}\right) \pm 1 \mp \tanh^2\left(\frac{t}{\sqrt{2}}\right) \right]$$

$$= \tanh\left(\frac{t}{\sqrt{2}}\right) \left[\mp \operatorname{sech}^2\left(\frac{t}{\sqrt{2}}\right) \pm \operatorname{sech}^2\left(\frac{t}{\sqrt{2}}\right) \right] = 0 \quad \square$$

* $q(t) = \pm \tanh\left(\frac{t}{\sqrt{2}}\right)$ is indeed a solution of $\ddot{q} + q - q^3 = 0$



Consider a particle near the surface which is constrained to move on frictionless surface defined as:

$$z = -\frac{\alpha}{\rho} + \beta \quad \text{where} \quad \rho = \sqrt{x^2 + y^2}$$

$$L = T - U = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

It's tempting to work in cylindrical coordinates with:

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z \Rightarrow \begin{aligned} \dot{x} &= \dot{\rho} \cos \phi - \rho \sin \phi \dot{\phi} \\ \dot{y} &= \dot{\rho} \sin \phi + \rho \cos \phi \dot{\phi} \end{aligned}$$

Also letting $z = f(\rho)$, our Lagrangian becomes:

$$L = \frac{1}{2} m \left[\dot{\rho}^2 + \rho^2 \dot{\phi}^2 + \dot{z}^2 \left(\frac{\partial f}{\partial \rho} \right)^2 \right] - mg f(\rho)$$

$$V(\rho) = -\frac{GMm}{\rho} = mg f(\rho) \Rightarrow f(\rho) = -\frac{GM}{g} \frac{1}{\rho} = -\frac{\lambda^2}{\rho} \quad \text{where} \quad \lambda = \sqrt{\frac{GM}{g}}$$

$$L = \frac{1}{2} m \left\{ \left[1 + \left(\frac{\partial f}{\partial \rho} \right)^2 \right] \dot{\rho}^2 + \rho^2 \dot{\phi}^2 \right\} - V(\rho) \rightarrow \text{gravitational potential}$$

If we consider Kepler's law for objects moving on a plane under the influence of a gravitational potential field:

$$L = \underbrace{\frac{1}{2} m \dot{\rho}^2}_{\text{radial velocity}} + \underbrace{\frac{1}{2} m (\rho \dot{\phi})^2}_{\text{tangential velocity}} - \underbrace{V(\rho)}_{-\frac{GMm}{\rho}} = \frac{1}{2} m \left\{ \dot{\rho}^2 + \rho^2 \dot{\phi}^2 \right\} - V(\rho)$$

Genericly, lagrangian of our toy simulator is not the same with the lagrangian for kepler's problem. BUT

if we have $\frac{\partial f}{\partial p} \ll 1$ we would obtain the same Lagrangians.

$$\frac{\partial f}{\partial p} = \frac{\partial z}{\partial p} = \frac{\partial}{\partial p} \left(\frac{-\alpha}{p} + \beta \right) = \frac{\partial}{\partial p} \left(-\frac{\lambda^2}{p} \right) \Rightarrow \frac{\lambda^2}{p^2} \ll 1 \text{ or } \frac{\alpha}{p^2} \ll 1$$

So what we essentially require is that change of z to be as close to zero as possible. This means if an objects move in a circular orbit $\frac{\partial f}{\partial p} = 1$ since p & z are constant throughout the motion. And toy can perfectly simulate the motion of an objects bound to kepler's laws for an exactly circular orbit.

Close to circular orbits $\epsilon \approx 0$ can also simulate the kepler's laws up to a certain precision.

However, if we have an object moving on an ecentric orbit ($\epsilon \approx 1$), our simulator fails since z changes greatly with p . In another words, orbit fails to have $p \gg \sqrt{\frac{GM}{g}}$

One can envision a contraption where an elastic fabric is stretched over and a large mass is placed over the fabric, effectively bending the fabric. Then one throws a marble and observes it's motion. If marble is far away from the mass and it's been given a tangential initial speed, it will aproximately move in a circular orbit, conserving it's z coordinate. But if it's initially moving towards the mass, it will spiral around the mass after gradually coming to a stop rather than following a very ecentric orbit.

In order to find the constraint force we should use Lagrange multipliers technique with $f(z, p) = z + \frac{\alpha}{p} - \beta = 0$

$$L^\lambda = T - U + \lambda f(z, p) \\ = \frac{1}{2} m (p^2 \dot{\phi}^2 + \dot{p}^2 + \dot{z}^2) - mgz + \lambda \left(z + \frac{\alpha}{p} - \beta \right) = 0$$

If Euler-Lagrange equations hold we have:

$$\text{For } p : \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{p}} \right) = m \ddot{p} = \left(\frac{\partial L}{\partial p} \right) = mp \dot{\phi}^2 - \lambda \frac{\alpha}{p^2}$$

$$\text{For } \dot{\phi} : \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = \frac{d}{dt} (mp^2 \dot{\phi}) = \frac{\partial L}{\partial \phi} = 0 \Rightarrow mp^2 \dot{\phi} = l : \text{const}$$

$$\text{For } z : \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{z}} \right) = m \ddot{z} = -mg + \lambda$$

$$\text{For } \lambda : 0 = z + \frac{\alpha}{p} - \beta \Rightarrow \ddot{z} = \frac{2\alpha}{p^3} \ddot{p}$$

$$m \ddot{z} = \frac{2m\alpha}{p^3} \ddot{p} = \frac{2\alpha}{p^3} (mp \dot{\phi}^2 - \lambda \frac{\alpha}{p^2}) = -mg + \lambda$$

But if we have $\frac{\partial f}{\partial z} \ll 1$, $\ddot{z} \approx 0 \Rightarrow$ we're on a close to circle ellipse

Then $\lambda = mg$

$$F_i = \sum_k \lambda \frac{\partial f_k}{\partial x_i} \Rightarrow \begin{aligned} F_z &= mg \frac{\partial f}{\partial z} = mg \\ F_p &= mg, \frac{\partial f}{\partial p} = -\frac{mg\alpha}{p^2} = -\frac{GMm}{p^2} \quad \text{as expected.} \end{aligned}$$

$\hookrightarrow$ Signs can be deceiving and one should not really care about them